Unconfounded but Inflated Causal Estimates

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Abstract

Quasi-experimental studies make empirical economics credible. To avoid confounding, causal identification strategies focus on a subset of the variation in the data, the plausibly exogenous part. In this paper, I argue that it can reduce statistical power and lead published estimates to exaggerate true effects sizes. Causal strategies avoid a bias at the cost of generating another one. I show the existence of this trade-off between confounding and exaggeration resulting from the use of causal inference methods through realistic fake data simulations and a mathematical derivation. I then discuss potential avenues to address this issue.

Preliminary introduction

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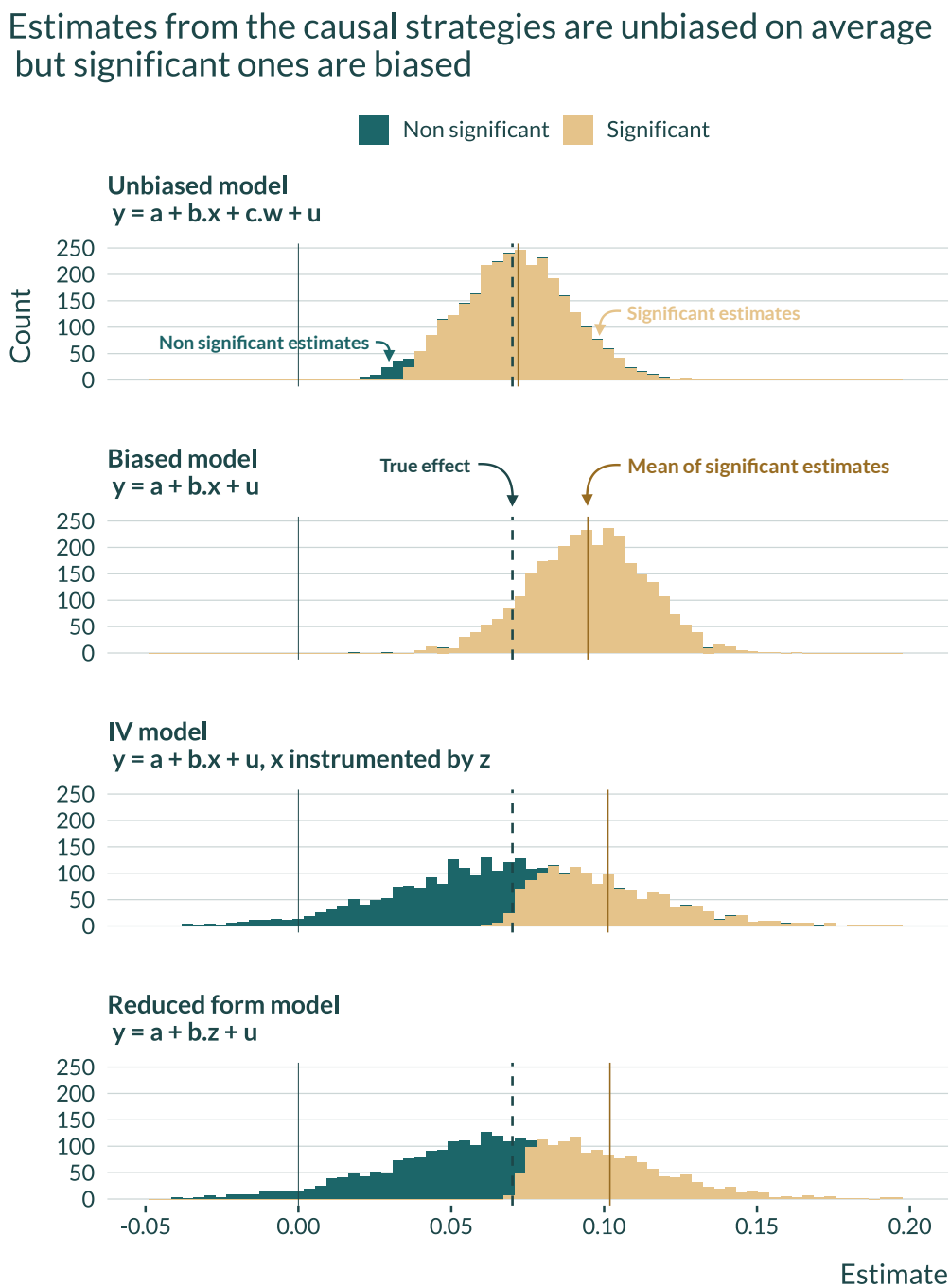
I Introduction

One of the main challenges of empirical economics is to reduce confounding to identify causal effects. Identifications strategies such as Regression Discontinuity (RD), Instrumental Variable (IV), Difference-in-Differences (DID) and event studies help achieve this goal. To do so, these strategies only use part of the variation in the data. They exploit the exogenous part of the variation in the treatment or decrease the sample size by only considering observations for which the as-if random assignment assumption is credible. Reducing the variation used decreases precision and thus statistical power, that is to say the probability of rejecting the null hypothesis of no effect when it is false. In this paper, I argue that it often creates a tension between statistical power and reducing confounding.

When statistical power is low, not only the estimator is imprecise as we well know but statistically significant estimates exaggerate the true effect size (Ioannidis 2008, Gelman and Carlin 2014, Lu et al. 2019, Zwet and Cator 2021). Only estimates at least two standard errors away from zero are statistically significant at the 5% level. In under-powered studies, these estimates make up a selected sub-sample of all estimates, located in the tails of the distribution of all possible estimates. The average of these statistically significant estimates differs from the true effect, located at the center of the distribution if the estimator is unbiased. When power is low, obtaining a statistically significant estimate from an unbiased estimator does not guarantee that it will be close to the true effect. An estimate $\hat{\beta}$ of β might be unbiased in the traditional sense but conditionally biased in the sense that $\mathbb{E}[\hat{\beta}] = \beta$ but $\mathbb{E}[\hat{\beta}|\text{ Significant}] \neq \beta$. Figure 1 illustrates this trade-off through the estimation of different models on simple simulated datasets.

This consequence of low statistical power could be non-problematic if a large literature had not underlined the existence of a publication bias favoring statistically significant results (Rosenthal 1979, Andrews and Kasy 2019, Abadie 2020, Brodeur et al. 2020, for instance). Published estimates from under-powered studies thus form a biased sample of the distribution of causal estimates and can greatly exaggerate their true effect sizes. This participates to the current replication crisis affecting various fields such as economics, epidemiology, medicine or psychology (Button et al. 2013, Open Science Collaboration 2015, Camerer et al. 2016, Chang and Li 2022). Even in experimental economics, with a high level of control and an arguable absence of confounders, estimates published in top economic journals were on average inflated by a factor of at least 1.5 (Camerer et al. 2016). Quasi-experimental studies are likely even more exposed to this exaggeration issue as, in current practices, statistical power is not central to the analysis. Despite usually large sample sizes, Ioannidis et al. (2017) concernedly finds that the median statistical power in a wide range of economic studies is no more than 18% and that nearly 80% of estimates may be exaggerated by a factor of two. The magnitude of these exaggerations is thus considerable and in some situations,

Figure 1 – Distribution of estimates across models on simple simulated datasets



Notes: y, x, w and u are respectively the outcome variable, the explanatory variable of interest, an unobserved variable and an error term. Each model is estimated on 3000 simulated datasets. Parameter values are chosen to produce designs with low power. Parameter values are set to make the IV and reduced form estimands comparable (an increase in z yields the same increase in x).

it could be on par with that of bias caused by confounders. It is thus crucial to take exaggeration risks into account.

In this paper, I show that design choices in quasi-experimental studies can be seen as a trade-off between avoiding confounding and overestimating true effect sizes due to a resulting loss in power. To limit the threat of confounding, causal inference methods discard variation and therefore reduce statistical power. This can result in inflated significant and published estimates. In some settings, reaching unbiasedness can actually be impossible as getting rid of bias caused by confounders creates another type of bias.

While the general idea that causal inference methods discard variation to identify effects is shared across strategies, the confounding / exaggeration trade-off is mediated through a distinctive channel for each of them. In RD designs, while the initial sample size may be large, we discard part of the variation by only considering observations within the bandwidth, decreasing the effective sample size and thus precision. In an IV setting, we only use the subset of the variation in the treatment that is explained by the instrument. In event studies, the variation used to identify an effect sometimes only comes from a limited number of treated observations. Assuming that all confounders are measured, matching prunes treated units that cannot be matched to untreated ones. Generic controls or fixed effects can increase the variance of the estimate if they absorb more of the variation in the treatment than in the outcome variable.

Since causal identification strategies can be interpreted as ways of controlling for confounders, this last point actually ties all the strategy specific arguments together. We implement Fixed Effects (FEs) based identification strategies such as DiD to control for the invariant, unobserved and arguably endogenous part of the variation in the outcome. In the Control Function (CF) approach of IV, we control for the predicted residuals of the regression of the endogenous variable of interest on the instrument. Fuzzy-RD and propensity score matching can also be thought of as control function approaches, of the forcing variable and propensity score respectively. In addition, excluding observations that are outside the bandwidth or unmatched is equivalent to controlling for observation level fixed effects for these observations. Considering a simple linear model gives the intuition for the trade-off between exaggeration and omitted variable bias when controlling. Let $y_i = \alpha + \beta x_i + \delta w_i + u_i$, $\forall i \in \{1, .., n\}$, with w potentially unobserved. Using the Frisch-Waugh-Lovell theorem we get that σ^2_{OVB} and σ^2_{CTRL}, the variance of the estimators for β when omitting w controlling for it respectively are:

$$\sigma^2_{\text{OVB}} = \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma_x} \qquad \text{and} \qquad \sigma^2_{\text{CTRL}} = \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma_{x^{\perp w}}}$$

and thus,

$$\sigma^2_{\text{CTRL}} > \sigma^2_{\text{OVB}} \quad \Leftrightarrow \quad \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma_{x^{\perp w}}} > \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma_x} \quad \Leftrightarrow \quad \frac{\sigma^2_{y^{\perp x,w}}}{\sigma_{y^{\perp x}}} > \frac{\sigma^2_{x^{\perp w}}}{\sigma_x}$$

Controlling will increase the variance of the estimator if the fraction of the variance unexplained by w is greater for $y^{\perp x}$ than for x. Put differently, if controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), it will increase the variance of the estimator. As briefly discussed above, since exaggeration increases with the variance of the estimator, controlling may increase exaggeration. I develop this intuition and a formal proof showing that exaggeration can be larger when controlling, even when accounting for bias, in section III.

In the remainder of the paper, I first illustrate the existence of this "causal exaggeration" using examples drawn from education, labor, environmental and political economics. The inflation of statistically significant estimates can be defined as the ratio of the estimated effect over the true effect, which is never known in real world setting. I therefore have to turn to fake-data simulations. Since these Monte-Carlo simulations have an illustrative purpose only, I intentionally focus on settings in which statistical power can be low. All other simulation assumptions are chosen to make it easier to retrieve the effect of interest. I consider simple linear models with constant treatment effects and all the models are correctly specified and accurately represent the data generating process, except for the omitted variable.

Then, I develop the mathematical intuition behind the confounding / exaggeration trade-off through the tying interpretation of causal identification strategies as control strategies. I then derive a formal proof of the existence of the trade-off in simple linear homoskedastic settings, with an homogeneous treatment effect. Specifically, I show that the bias caused by exaggeration can be larger than the one caused by confounders, accounting for omitted variable bias. I also study the drivers of exaggeration and show that it increases as the strength of the instrument decreases, the number of exogenous shocks decreases or when controlling for an omitted variable absorbs more of the variation in the treatment than in the outcome.

Finally, I discuss avenues to address this causal exaggeration. First, I advocate for the use of tools to evaluate the potential magnitude and risk of both confounding and exaggeration issues separately. Sensitivity analyses help with the former while power calculations help with the latter. By approximating the data generating process, prospective power simulations help identify the design parameters affecting power and exaggeration (Gelman 2020, Black et al. 2021). Retrospective power calculations allow to evaluate whether a study would have enough power to confidently estimate a range of smaller but credible effect sizes (Gelman and Carlin 2014, Stommes et al. 2021). The companion website describes in details how such solutions can be implemented. Focusing more specifically on the trade-off and its drivers, I present tools to visualize the variation actually used for identification when using causal identification strategies. Finally, I briefly discuss

potential solutions to mitigate this trade-off.

This paper contributes to three strands of the literature. First, the idea that causal identification estimators, while unbiased, may be imprecise is not new; this is part of the well-known bias /variance trade-off (Imbens and Kalyanaraman 2012, Deaton and Cartwright 2018, Hernán and Robins 2020, Ravallion 2020). In under-powered studies, resulting estimates have large confidence intervals, suggesting that a wide range of effects are consistent with the data. I approach this literature from a different angle: through the prism of statistical power and publication bias. Not only the limited precision resulting from the use of causal identification methods could make it difficult to draw clear conclusions regarding the exact magnitude of the effect but I argue that it might also inherently lead to inflated published effect sizes, creating another "bias".

Second, recent studies discussing the inflation of statistically significant estimates due to low power focused on specific causal identification methods separately and usually do not investigate the determinants of this exaggeration (Schell et al. 2018, Black et al. 2021, Stommes et al. 2021, Young 2021). In this paper, I propose an overarching mechanism, inherent to causal identification strategies as a whole, that can explain these issues: although each strategy does so through a different means, in essence they discard part of the variation, thereby causing exaggeration. This connection could be exacerbated by the fact that, as noted by Brodeur et al. (2020), publication bias is more prevalent for some methods such as the IV.

Third, this study contributes to the literature on reproducibility in economics (Camerer et al. 2016, Ioannidis et al. 2017, Christensen and Miguel 2018, Kasy 2021). The trade-off presented in this paper may be an additional explanation for observing replication failures in empirical economics, despite the widespread use of convincing causal identification methods.

II Practical Recommendations

We have seen that using causal methods induces a trade-off between avoiding confoundings and exaggerating true effects if the study is under-powered. How can we, as applied researchers using observational data, arbitrate it? Since key pieces of information such as the true effect and the effect of omitted variables are inherently unknown, we cannot directly compute the biases caused by confounders and exaggeration. In this section, I examine how we however can get a sense of threats from both sides of the trade-off and visualize its main driver, the variation used for identification. I then discuss how changing our attitude towards statistical significance and replicating studies could limit the exaggeration issue.

II. 1 Gauging omitted variable bias

On one side of the trade-off highlighted in this paper lies the bias caused by confounders. Although it is in essence impossible to measure, tools such as sensitivity analyses are available to gauge its magnitude (Rosenbaum 2002, Oster 2019, Cinelli and Hazlett 2020). For instance, the method developed in Cinelli and Hazlett (2020) enables to assess how strong confounders would have to be to change the estimate of the treatment effect beyond a given level we are interested in. It offers bounds for the strength of the association between the treatment and potential omitted variables by weighting it against the measured association between the treatment and observed covariates. A typical conclusion from such an analysis would be: "omitted variables would have to explain as much residual variance of the outcome and the treatment as this observed covariate to bring down the estimate to a value of X". In addition, the authors implement graphical tools to carry out this comparison. I propose to use such quantitative bias analyses to evaluate whether the threat of unobserved confounding requires a restrictive causal approach.

II. 2 Evaluating risks of exaggeration

On the other side of the trade-off lies the exaggeration emerging when statistical power is low. As OVB, exaggeration and statistical power are in essence impossible to measure as their computation depends on the true effect which is always unknown. Yet, power calculations can be computed by making hypothesizes on the magnitude of the true effect. In randomized controlled trials, such computations are not only an established practice but a requirement (Duflo et al. 2007, McConnell and Vera-Hernandez 2015, Athey and Imbens 2016). They are however rarely reported in observational studies. Yet, the threat of exaggeration highlights the necessity of running power calculations in observational studies as well. A low power or a relatively large variance not only makes it more difficult to draw clear conclusions about the magnitude of the effect but it can also create a bias. I thus advocate to make power central to observational analyses and to run pre and post analysis power and exaggeration calculations. bb

There is a lack of guidance on how to design well-powered observational studies. In causal inference textbooks, very few pages are devoted to the topic (Angrist and Pischke 2009; 2014, Imbens and Rubin 2015, Cunningham 2021). To the best of my knowledge, only two textbooks discuss the matter in depth (Shadish et al. 2002, Huntington-Klein 2021).

Simulating the design of an observational study is a solution to overcome these limits (Hill 2011, Gelman 2020, Black et al. 2021). Similarly to what we did in the previous section, the goal of this approach is to simulate the data generating process of the study from scratch. It requires thinking about the distribution of the variables and their relationships. External information found in previous studies can help guide the simulation process to make it more realistic. If the

relationships among covariates are too complex to emulate, a second approach starts from an existing dataset to which a simulated treatment and potential outcomes are added [1].

When simulations indicate that statistical power is low, additional data could be collected or the statistical model could be expanded to increase precision. In any case, it should not stop from carrying out a research project. Simulation results rest on the way the data generation process was modeled and it can be difficult to gauge the amount of noise present in data before actually analyzing them. The two actual benefits of a prospective simulation procedure are to think about factors that affect power and not to be mislead by statistically significant estimates if power is low.

II. 3 Once the Main Analysis is Completed

Once we have obtained a statistically significant estimate for the treatment of interest, we still need to think about the statistical power of the study to check whether the magnitude of our estimate is trustworthy. A *retrospective* power analysis helps evaluate whether the design of the study would produce uninflated statistically significant estimates if the true effect was smaller than the observed estimate (Gelman and Carlin 2014, Ioannidis et al. 2017, Stommes et al. 2021).

We illustrate how a retrospective analysis works by taking the example of Card (1993) on the relationship between human capital and income. He finds that an additional year of education, instrumented by the distance of growing near a four-year college, causes a 13.2% average increase in wage. The associated standard error is 5.5%. As noted by the author himself, the estimate is very imprecise: if the true causal effect was slightly smaller than the observed estimate, the study would very likely be under-powered. For instance, imagine that prior evidence suggests that the true effect could be to closer a 10% increase in wage. Computing the statistical power of the study only requires to draw many estimates from a normal distribution centered around the hypothesized true effect of 10% and with a standard deviation equal to the 5.5% standard error obtained in Card (1993). Concretely, one proceeds as if they were able to replicate the study many times under the assumption that the true effect is different from the observed estimate. The proportion of sampled estimates that are statistically significant at the 5% level, 44% in this case, is the statistical power. The inflation of significant estimates is then computed as the average ratio of the values of statistically significant estimates over the assumed true effect size: these estimates would be 1.5 times too large on average.

For a retrospective power analysis to be useful, it is therefore necessary to make informed guesses about the range of plausible effect sizes. Such guesses can be based on results from meta-analyses or previous studies with a convincing design (e.g., a large randomized controlled trial).

1. In the future version of the paper, we will explain how we can easily simulate from scratch the study by Card (1993). For now, examples of simple simulations are available on our companion website.

When such information is not available, power calculations can be run for a range of smaller but credible effect sizes [2].

Results from power and exaggeration calculations would not only be highly informative but could also be reported very concisely in the robustness section of articles. `R` and `Stata` packages have been developed (Timm et al. 2019, Linden 2019) to easily implement retrospective power analyses.

II. 4 Attitude Towards Statistical Significance and Replication

General changes in scientific practices could also limit the inflation of statistically significant estimates in under-powered studies. As shown in our simulations, if estimates were not filtered by their statistical significance, even under-powered studies would on average recover the true effect. The publication bias arising from dichotomizing evidence according to p-values has long been criticized in many disciplines but has seen a revival with the recent replication crises in psychology, medicine and social sciences. Many researchers advocate abandoning statistical significance as a measure of a study's quality (McShane et al. 2019). This would essentially eliminate the trade-off described in this paper.

To be effective, this change in attitude towards statistical significance should be paired with an effort to replicate studies (Christensen and Miguel 2018). Replications, even of low powered studies, would eventually enable to build the distribution of the causal estimand of interest. Meta-analyzes would then reduce the uncertainty around the true value of the causal estimand by pooling estimates (Hernán 2021).

Finally, the inflation of statistically significant estimates can be limited by considering confidence intervals as compatibility intervals (Shadish et al. 2002, Amrhein et al. 2019, Romer 2020). The width of these intervals gives a range of effect sizes compatible with the data. Confidence intervals will be wide in under-powered studies signaling that point estimates should not be taken at face value, even if statistically significant.

2. We are currently working on the adoption of two existing approaches that could help address the potential inflation of statistically significant estimates. In a series of articles, Zwet and Gelman (2021), Zwet and Cator (2021) and Zwet et al. (2021) present a Bayesian procedure to shrink statistically significant estimates based on a corpus of estimates from prior studies. The second approach consists in carrying out quantitative bias analyses to evaluate whether the threat of unobserved confounding requires a restrictive causal approach. Rosenbaum (2002), Oster (2019) and Cinelli and Hazlett (2020) have developed different methods to run sensitivity analyses. They could be paired with power calculations to better evaluate the hidden bias / power trade-off of competing research designs. For instance, if a sensitivity analysis reveals that a simple selection on observables strategy is very robust to omitted variable bias, one may avoid using an IV model since it has a higher chance to produce inflated estimates.

III Conclusion

Causal identification strategies have undoubtedly participated in making empirical analyses more credible (Angrist and Pischke 2010). To avoid confounding, they only exploit the exogenous part of treatment variation. In this paper, we argue that the same aspect that makes causal identification strategies credible can create another type of bias. Not only the lack of precision makes it more difficult to precisely get a sense of the magnitude of the actual effect but it also increases the probability of published estimates to be inflated. The confounding / exaggeration trade-off we highlight in this paper manifests itself along different dimensions for each identification strategy. A systematic reporting of statistical power calculations in observational studies could help gauge the risk of falling into this low power trap.

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